DATA ANALYSIS · EXPLORATORY STUDY

What Really Drives the Price of a Used Car?

An end-to-end analysis of 5,500 cars, from raw data to insight

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The dataset at a glance

5,500	18	10	0	4
Used cars	Features per car	Brands, 40 models	Rows deleted	Viz libraries

About 15% of values were missing in 9 columns. They were filled using each model's own typical specs, so no row had to be deleted.

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The analysis journey

1

Audit

Missing values, duplicates and data types

2

Clean

Smart imputation, zero rows deleted

3

Explore

Distributions, correlations and group comparisons

4

Engineer

Age, yearly mileage and luxury tier

5

Conclude

Statistical tests separate signal from noise

From raw CSV to insight in five deliberate steps.

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Cleaning without losing data

Problem	Decision	Why
Missing specs: horsepower, torque, engine size, efficiency	Median per model and fuel type	Specs are stable within a model (horsepower spread 34 vs 56 overall)
Missing accident history	New “Unknown” category	Not knowing is not the same as no accident
Electric cars with engine size 0	Kept as is	Valid: they have no combustion engine
Extreme mileage and $500 prices	Kept and flagged	Real market behavior, not errors

About 15% of values were missing in 9 columns, and the gaps were random (max correlation 0.033), so imputation is safe.

Four libraries, one story

Matplotlib

Distributions and multi-panel layouts

Seaborn

Boxplots, heatmaps and grouped comparisons

Plotly

Interactive scatter, treemap, parallel coordinates and a US map

Altair

Linked dashboard: select a region and watch other charts update

Pandas, NumPy and SciPy power the cleaning, feature engineering and ANOVA tests behind every claim.

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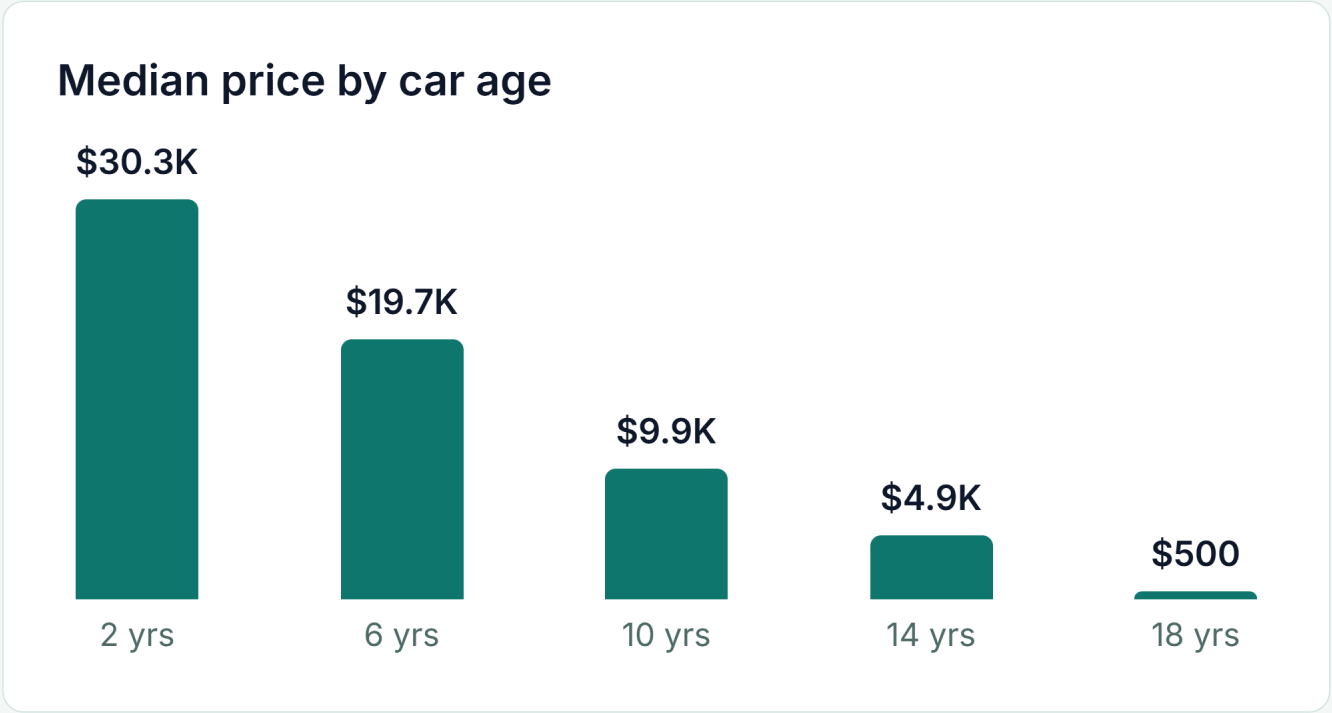
Age is the king of price

−0.80

correlation between age and price, the strongest in the data

≈14%

of a car's value is lost every single year



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A hidden price floor

22%

of cars sell at exactly $500

Older
average year 2008 vs 2016

Heavily driven
238K miles vs 79K

More owners
4 previous owners vs 2

Ownership hurts too: one owner, $22,846 median. Two owners, $10,764. More than half the value is gone at the first resale.

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The accident trap

Naive comparison

−75%

$14,119 without an accident vs $3,481 with one

Within the same age group

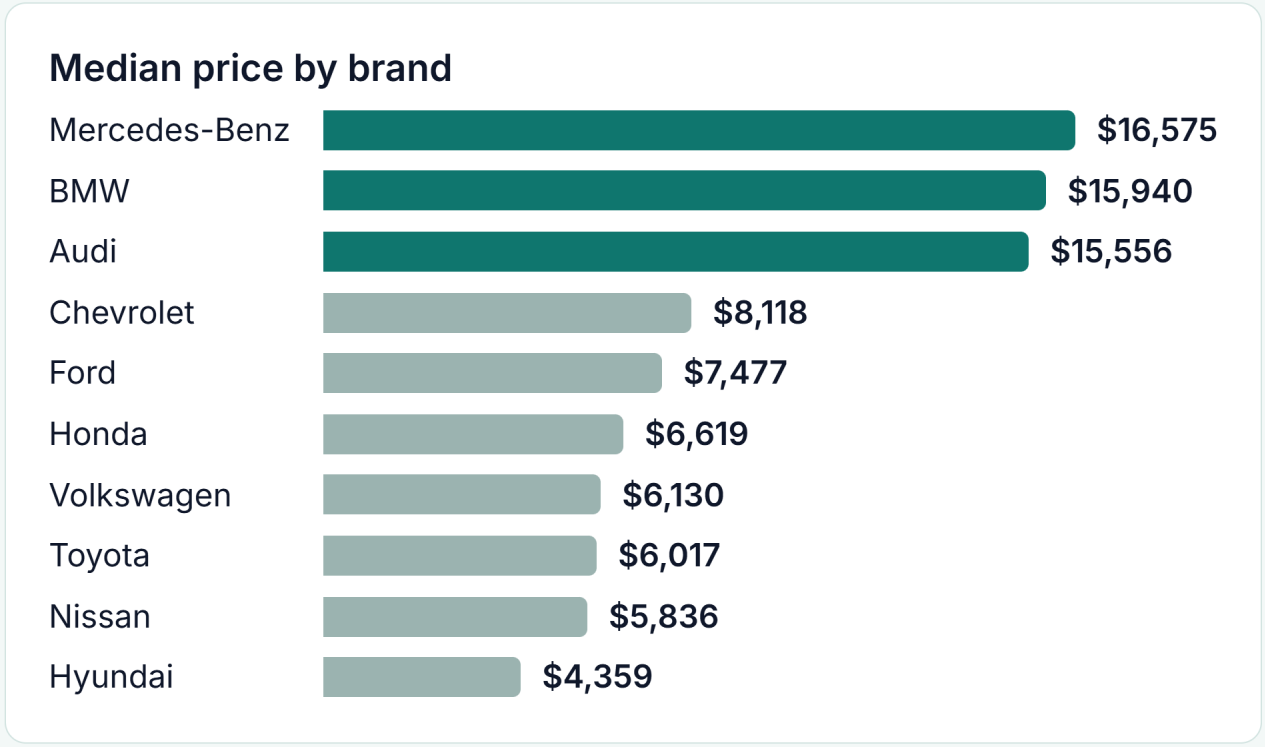
−18 to −26%

The real cost of an accident

Why the gap? Cars with accidents are simply older: 5% of cars aged 0–5 have one on record, versus 50% of cars aged 16+. Always control for age first.

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Two price tiers



$15.6K to $16.6K

median price of the three luxury brands

Under $8.2K

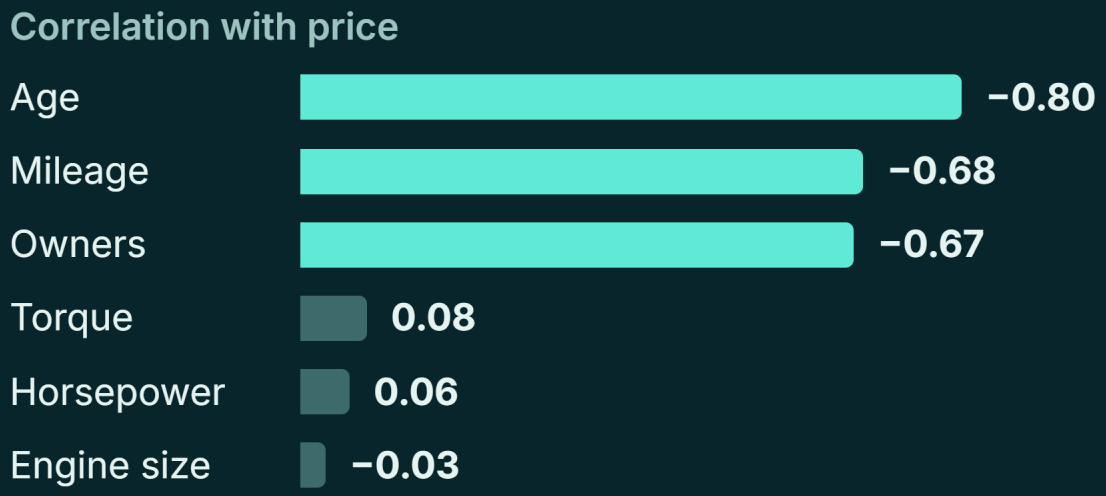
median price of every other brand

14.4% to 16.2%

yearly depreciation, similar across all brands

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What moves price, and what does not



Hidden link
Owners and age move together (0.83), so owners add little on their own.

Service history is real
Full service records sell for about $2,000 more (p = 0.007).

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Myth busters

Common belief	Verdict	What the data says
Electric and hybrid cars keep their value better	Busted	Same age, same price: $26,852 electric vs $26,111 petrol (0–5 yrs)
More horsepower means a pricier car	Busted	Correlation with price: only 0.06
Color and state change the price	Not proven	Differences are statistical noise (p = 0.16 and 0.44)
Luxury brands lose value more slowly	Barely	14.4% vs 16.2% lost per year across brands

Key takeaways

1
Age, mileage and owners decide the price

2
Control for age before drawing conclusions

3
Engine power, color and fuel type barely matter

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